Chapter 7

7.1.Try to determine unilateral Z transformation for the following periodic sequence.

(1) $x[k]=\begin{cases}1, & k=2n, \quad n=0,\ 1,\ 2,\cdots \\ 0, & k=2n+1, \quad n=0,\ 1,\ 2,\cdots\end{cases}$

(2) $y[k]=\sum_{i=0}^{k}(-1)^i x[k-i]$

7.2.We know $\mathscr{Z}\{x[k]\}=X(z)=\dfrac{1}{1+z^{-2}}, |z|>1$. By using the properties of Z transformation,try to obtain the following unilateral Z transformation and its convergence region.

(1) $x_1[k]=x[k-2]$ (2) $x_2[k]=(\frac{1}{2})^k x[k]$

(3) $x_3[k]=(\frac{1}{2})^k x[k-2]$ (4) $x_4[k]=kx[k]$

(5) $x_5[k]=(k-2)x[k]$ (6) $x_6[k]=\sum_{i=0}^{k} x[i]$

(7) $x_7[k]=\sum_{i=0}^{k} x[k-i]$ (8) $x_8[k]=\sum_{i=1}^{k} x[k-i]$

7.3.We know Z transformation of causality sequence $x[k]$ is $X(z)$,try to find the initial $x[0]$、$x[1]$ and final values $x[\infty]$ of $X[k]$.

(1) $X(z)=\dfrac{z(z+1)}{(z^2-1)(z+0.5)}$ (2) $X(z)=\dfrac{2z^2}{(z-\frac{1}{2})(z+\frac{1}{3})}$

7.4.Try to find the following unilateral Z inverse transformation using partial fraction method.

(1) $X(z)=\dfrac{z^{-2}}{2-z^{-1}-z^{-2}}$, $|z|>1$ (2) $X(z)=\dfrac{1-z^{-1}}{1+\frac{5}{4}z^{-1}+\frac{3}{8}z^{-2}}$, $|z|>\dfrac{3}{4}$

(3) $X(z)=\dfrac{1}{(1-z^{-1})(1+z^{-2})}$, $|z|>1$ (4) $X(z)=\dfrac{1-4z^{-1}}{(1-z^{-1})(1+5z^{-1}+6z^{-2})}$, $|z|>3$

(5) $X(z)=\dfrac{2}{z^2-z-2}$, $|z|>2$ (6) $X(z)=\dfrac{2z+1}{z^2+3z+2}$, $|z|>2$

(7) $X(z)=\dfrac{z}{(z^2+z+1)(z-1)}$, $|z|>1$ (8) $X(z)=\dfrac{z}{(z-1)^2(z+1)}$, $|z|>1$

7.5. Try to find zero input response, zero state response and complete response of the following causality discrete time LTI system.

(1) $y[k]-\frac{1}{3}y[k-1]=x[k]$, $x[k]=(\frac{1}{2})^k u[k]$, $y[-1]=1$;

(2) $y[k]-\frac{1}{3}y[k-1]=x[k]$, $x[k]=(\frac{1}{3})^k u[k]$, $y[-1]=1$;

(3) $y[k]-\frac{1}{3}y[k-1]=x[k]+x[k-1]$, $x[k]=(\frac{1}{2})^k u[k]$, $y[-1]=1$;

(4) $y[k]-\frac{4}{3}y[k-1]+\frac{1}{3}y[k-2]=x[k]$, $x[k]=(\frac{1}{2})^k u[k]$, $y[-1]=1$, $y[-2]=2$;

(5) $y[k]+y[k-1]-2y[k-2]=x[k-1]+2x[k-2]$, $x[k]=u[k]$, $y[-1]=-0.5$, $y[-2]=0.25$;

(6) $y[k]+y[k-2]=x[k]-x[k-2]$， $x[k]=(\frac{1}{2})^k u[k]$， $y[-1]=2$， $y[-2]=0$。

7.6. It is known that the step response of a discrete LTI system excited by step signal is $g[k]=(\frac{1}{2})^k u[k]$,try to determine

(1) The system function $H(z)$ and unit impulse response $h[k]$ of the system.

(2) Zero state response $y_{zs}[k]$ generated by $x[k]=(\frac{1}{3})^k u[k]$ excitation.

7.7. The difference equation describing a discrete LTI system is

$$y[k]+3y[k-1]+2y[k-2]=x[k]\ ,\qquad k\geq 0$$

And $x[k]=u[k],\quad y[-1]=-2,\quad y[-2]=3,$ try to determine by Z domain.

(1) Zero input response $y_{zi}[k]$, zero state response $y_{zs}[k]$, complete response $y[k]$.

(2) System function $H(z)$, unit impulse response $h[k]$.

(3) If $x[k]=u[k]-u[k-5]$, resolve (1)、(2).

7.8. A causality discrete LTI system with an initial state of zero,when input is $x[k]=u[k]$,output is

$$y[k]=[(\frac{1}{2})^k-(\frac{1}{3})^k+2]u[k]$$

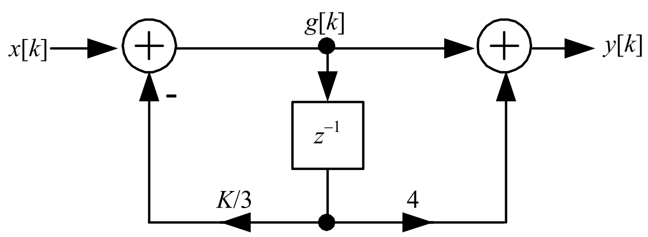
Try to determine the difference equation describing the system.

7.9. Try to draw a direct form of the following discrete time system, cascaded and parallel simulation block diagram.

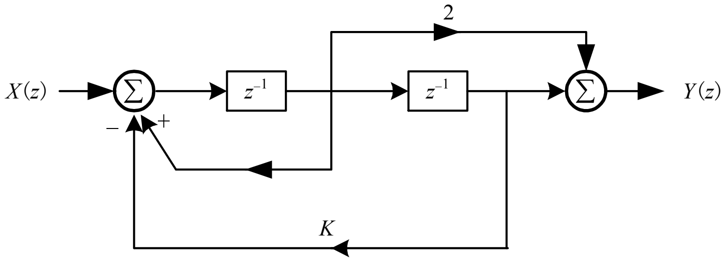
(1) $H(z)=\dfrac{1+z^{-2}}{(1+\frac{1}{2}z^{-1})(1-\frac{1}{3}z^{-1})}$ (2) $H(z)=\dfrac{1+2z^{-1}}{(1-z^{-1}+\frac{1}{2}z^{-2})(1-\frac{1}{2}z^{-1})}$

(3) $H(z)=\dfrac{2z+1}{z(z-1)(z-0.5)^2}$ (4) $H(z)=\dfrac{z(z+2)}{(z-0.8)(z+0.4)(z-0.6)}$

7.10. The simulation block diagram of discrete causal system is shown in Figure 7-1. Find the system function $H(z)$ and determine the K value range when the system is stable.



(a)



(b)

Figure.7-1